# Details on the angle between two lines and the slops

Let the two lines make angles $\alpha$ and $\beta$ with the positive $x$-axis. Their slopes are

$$m_1 = \tan\alpha, \qquad m_2 = \tan\beta.$$

The angle between the lines is $\theta = |\alpha - \beta|$ (taking the smaller angle). The tangent subtraction identity says

$$\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}.$$

Replacing $\tan\alpha$ and $\tan\beta$ by the slopes gives

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|.$$

The absolute value ensures the angle is nonnegative. If $1 + m_1 m_2 = 0$, the lines are perpendicular, so $\theta = 90^\circ$ and $\tan\theta$ is undefined.

For the **homogeneous pair**

$$ax^2 + 2hxy + by^2 = 0,$$

write each line through the origin as $y = mx$. Substitution gives

$$bm^2 + 2hm + a = 0.$$

Its roots $m_1, m_2$ are the slopes of the lines. Using

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

and the sum and product of the roots gives

$$\boxed{\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}} \quad \text{when } a + b \neq 0.$$

If $a + b = 0$, the lines are perpendicular, so $\theta = 90^\circ$. This applies to two distinct real lines, for which $h^2 > ab$.



# General Equation of Second degree (Class work)

**Q**. Find the condition that the general equation of the 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ may represent a pair of straight lines. Find point of intersection.

**Solution**: Given $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$……..(1)

If the origin transferred to the point $(\alpha, \beta)$

Put $x = \alpha + x'$

and $y = \beta + y'$

Form equation (1) we get $ax'^2 + 2hx'y' + by'^2 + 2(a\alpha + h\beta + g)x' + 2(h\alpha + b\beta + f)y' + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$……(2)

The equation (1) may represent a pair of straight lines, if it is reduced to a homogeneous equation in $x$ and $y$. This is possible if the coefficient of $x$ and $y$ and the constant term equal to zero

$a\alpha + h\beta + g = 0$……..(3)

$h\alpha + b\beta + f = 0$……(4)

and $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$

$\Rightarrow a\alpha^2 + h\alpha\beta + g\alpha + b\beta^2 + h\alpha\beta + f\beta + g\alpha + f\beta + c = 0$

$\Rightarrow \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c = 0$

$\Rightarrow \alpha.0 + \beta.0 + g\alpha + f\beta + c = 0$

$\therefore g\alpha + f\beta + c = 0$……(5)



# General Equation of Second degree (Class work)

From (3), (4) and (5) we get

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

Point of intersection: $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\Rightarrow \frac{\partial}{\partial x}(ax^2 + 2hxy + by^2 + 2gx + 2fy + c) = \frac{\partial}{\partial x}(0)$$

$$\Rightarrow 2ax + 2hy + 2g = 0$$

$$\therefore ax + hy + g = 0 \text{……(6)}$$

Again $\frac{\partial}{\partial y}(ax^2 + 2hxy + by^2 + 2gx + 2fy + c) = \frac{\partial}{\partial y}(0)$

$\Rightarrow 2hx + 2by + 2f = 0$

$\therefore hx + by + f = 0$……..(7)

Solving (6) and (7) we get

$x = \frac{hf - bg}{ab - h^2}$ and $y = \frac{hg - af}{ab - h^2}$

## Exercise (HW)

Prove that $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represent a pair of straight. Find also their point of intersection and angles between them.

Solution: Given $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$

Now compare with general equation of 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = -3, h = -4, b = 3, g = -\frac{29}{2}, f = \frac{3}{2}$ and $c = -18$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

We have $\Delta = 0$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b} \Rightarrow \theta = 90^0$$

$$x = \frac{hf-bg}{ab-h^2} \text{ and } y = \frac{hg-af}{ab-h^2}$$

After solving we get

$x = 3/2$ and $y = 5/2$

So the required point is $(x, y) = (\frac{3}{2}, \frac{5}{2})$



*Thanks for your attention*